

University of Camerino
MSc in Computer Science, Artificial Intelligence curriculum · Complex Systems Design

Intelligent Search Across Organizational Knowledge

AI Puzzle piece P14 · Technical report

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Abstract

People inside an organization look for rules and procedures with their own everyday words, while documents use formal and legal language. Keyword search fails in this case. This project builds **P14**, a local search service that finds information **by meaning and by keyword**, always returns the **exact source** of every passage, filters results by **access permission**, and says so when it does not have enough evidence.

The case study is a fictional public body, "Ente Demo": 32 real Italian official acts plus 25 synthetic internal documents, for a total of 57 documents and 10,689 indexed passages. On a frozen test set of 70 queries, hybrid search (S2) reaches nDCG@10 = 0.604 and Recall@10 = 0.760, against 0.453 and 0.557 for the keyword baseline (B0); the difference is statistically clear ($p \leq 0.0002$). No restricted passage was returned to an unauthorised user in 18,576 checks, and the median response time is under 50 ms. One of our own targets was not met: the correct source appears in the top 3 for 76.7% of the queries, below the 80% we set.

The service exposes a documented REST/OpenAPI contract, so that other pieces of the AI Puzzle (P24, P12, P11, P15, P25) can use it. Two of these connections are implemented as working mock consumers.

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1. Introduction and AI Puzzle context

The course uses the **AI Puzzle framework** (consolidated version 5.0): 26 autonomous AI projects for public and private organizations. Each group builds one piece. A piece must be useful on its own and, at the same time, prepared so that its outputs can become the inputs of other pieces. Only in the final stage of the course are the pieces connected, so that the behaviour of the complex system emerges from their interaction.

P14 belongs to the area *Documents, Knowledge & Organizational Memory*. It sits low in the functional stack: it turns documents into verifiable evidence that other pieces can consume. For this reason two properties are central in the whole design:

- **Provenance**: every output must point to its exact source.
- **Interfaces**: other pieces need a stable contract and persistent identifiers.

The assignment of P14 to this project is provisional and awaits confirmation from the professor.

2. Problem and objectives

The P14 project sheet asks for a search engine that finds information by meaning, not only by exact keyword match, across heterogeneous documents, and that always returns the precise references to its sources. The four objectives of the sheet and how this project addresses them:

Table 1. Objectives of the P14 sheet and how they are covered.

Objective (P14 sheet)	How it is addressed
Reduce the time needed to find information	Ranked passages instead of whole documents; median latency under 50 ms (measured only through proxies, no user study)
Retrieve content even with different wording	Dense retrieval with a multilingual Transformer, fused with BM25; measured separately on paraphrased and English queries
Connect relevant sources on the same topic	Results mix laws and internal rules on the same topic; a dedicated "related documents" endpoint is future work
Always make provenance verifiable	Every passage carries document, version, section, page and character offsets, and can be resolved again by its id

Case study. There is no real organization available, so the project uses a fictional public body, **Ente Demo**. Its knowledge is the Italian legislation it must apply plus its own internal documents. Crawling a real intranet or mailbox and any free-text generated answer are out of scope.

3. Requirements traceability

The framework lists eight common deliverables for every group. Table 3 shows where each one is satisfied in the repository.

Table 2. The eight common deliverables of the AI Puzzle framework.

#	Common deliverable	Where in the repository
1	Architecture and rationale of design choices	docs/ARCHITECTURE.md, docs/diagrams/, docs/DECISIONS.md, Section 4
2	Executable prototype on the case study	src/p14_search (API), frontend/app.py (demo), scripts/, README.md
3	Data description: sources, schema, quality, limits, privacy	docs/DATA.md, Section 5

#	Common deliverable	Where in the repository
4	Experimental evaluation with baseline, metrics, success criteria	docs/EVALUATION.md, eval/, eval/results/, Section 7
5	Contract for the integration interface	contracts/openapi/p14-search.yaml, contracts/schemas/, docs/INTEGRATION.md
6	One real input and one real machine-readable output	contracts/examples/IngestDocument.real.json, SearchResponse.real.json
7	At least two connections with other pieces	P24 and P12 implemented as mocks; P11, P15, P25 documented (Section 9)
8	Human oversight, traceability, security, validation before automation	docs/SECURITY_OVERSIGHT.md, Section 10

4. Architecture and design decisions

The system is deliberately simple: **one FastAPI service, one Qdrant container and one SQLite file**, all running locally on a laptop, with no microservices and no paid service. Figure 1 shows P14 among its users and the other puzzle pieces; Figure 2 shows its internal modules.

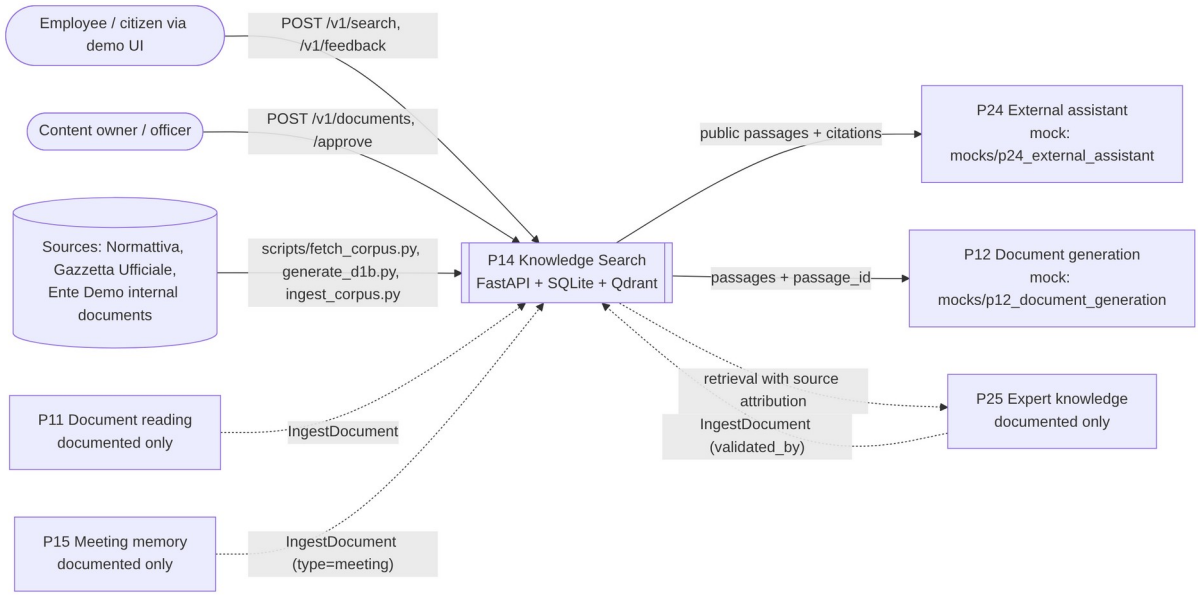


Figure 1. System context: P14 between its users, its data sources and the other AI Puzzle pieces. Solid lines are implemented; dotted lines are documented.

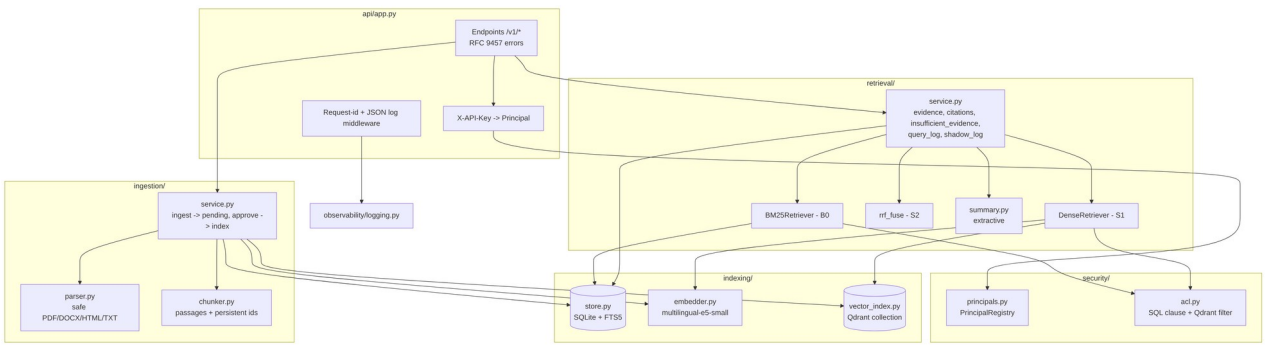


Figure 2. Internal components of the service (module names as in src/p14_search/).

4.1 Main design decisions

Table 3. Main design decisions (full records in docs/DECISIONS.md).

Decision	Choice	Why
Retrieval method	Hybrid: BM25 + dense, merged with Reciprocal Rank Fusion (RRF, k = 60)	BM25 finds exact legal terms and article numbers; the dense model finds meaning. RRF uses only ranks, so the two score scales do not need to be normalised.
Keyword index	SQLite FTS5 (ADR-004)	Already in the stack, persistent, permission filters are a simple SQL WHERE clause, no extra service.
Embedding model	intfloat/multilingual-e5-small (ADR-005)	Small, runs on an 8 GB GPU, handles Italian and English; it beat the alternative on the tuning split only.
Access control	Applied inside both retrievers, not after ranking	Unauthorised passages never enter ranking, so they cannot leak through counts or scores.
Human approval	A document is indexed only after its owner approves it (ADR-007)	A person decides what becomes searchable and at which access level.
Identifiers	document_id from the source; version = content hash; passage_id = hash(document, version, chunker, offset)	Citations stay valid; other pieces can store and resolve passage ids later.
Scores	Exposed as "evidence", not "confidence"	Retrieval scores are not calibrated probabilities.
Parsing	Safe parser with type and size checks (ADR-006)	Rejects macros, external entities and zip bombs; handles difficult official PDFs.

5. Data

The knowledge base of Ente Demo combines real public legislation (D1a) with synthetic internal documents (D1b). A public benchmark (D2) is used only to check that the pipeline behaves as expected.

Table 4. Datasets used in the project (statistics from eval/results/corpus_stats.json).

Dataset	What it is	Size	Licence and limits
D1a · real	Italian official acts (workplace safety, smart working, privacy, digital administration, transparency, whistleblowing…) from Normattiva and the Gazzetta Ufficiale	32 documents, 10,561 passages	Official acts are excluded from copyright (art. 5, L. 633/1941). GU issues also contain unrelated acts and some personal data.
D1b · synthetic	Internal regulations, procedures, FAQs, circulars, a meeting minute, a budget of Ente Demo; public, internal and restricted	25 documents, 128 passages	Written by an LLM and labelled as synthetic; no personal data; legal references not verified.
D2 · benchmark	BEIR SciFact: scientific claims and abstracts with independent relevance labels	5,183 abstracts, 300 test queries	CC BY 4.0 / ODC-By 1.0; English, different domain: sanity check only.

In total the case study has **57 documents and 10,689 passages** (mean passage length about 670 characters). The internal documents are only 0.5% of the text: internal knowledge is a needle in a haystack of legislation, which is realistic and makes the access-control tests meaningful. Data quality

was a real issue: amendment markup in HTML, PDF fonts without a Unicode map, and unrelated acts inside Gazzetta issues. These problems and their handling are documented in docs/DATA.md.

6. Implementation

The code is written in Python 3.11 and organised in modules: `api`, `ingestion`, `indexing`, `retrieval`, `security`, `evaluation`, `observability`. 80 automated tests cover parser safety, identifiers, API behaviour, access-control leakage, contract conformance, the mock consumers and the demo interface; all 80 pass.

6.1 Ingestion

A document is validated, parsed safely, split into passages with persistent ids and stored as *pending*. Only when its owner approves it, with an access level, is it embedded and written to both indexes (Figure 3).

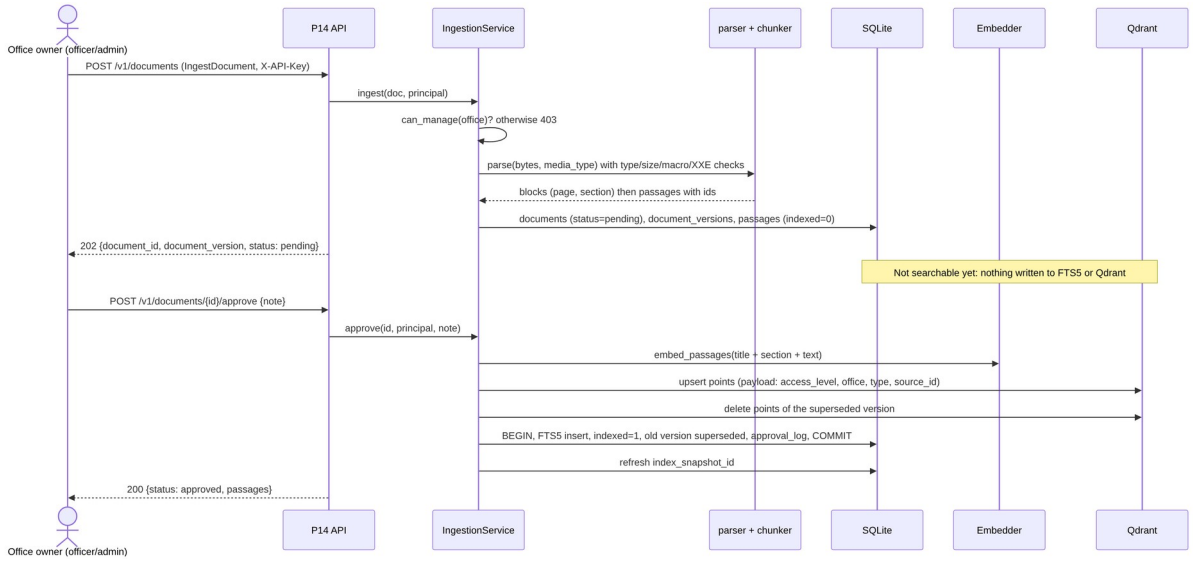


Figure 3. Ingestion with human approval.

6.2 Search

For each query the API key is turned into a principal (role, access level, offices). Keyword and dense retrieval run with the permission filter inside each index; their rankings are fused; passages get their citations; an extractive summary is built only from returned sentences; the evidence check decides whether to raise `insufficient_evidence`; finally the request is logged (Figure 4).

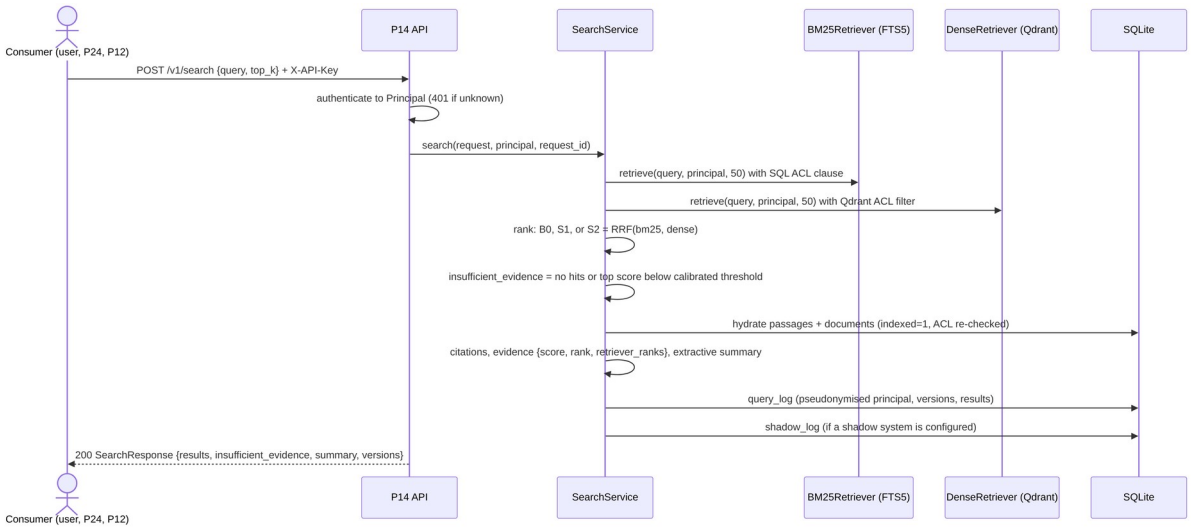


Figure 4. Search with permission filter, fusion, citations and logging.

6.3 Classes and data model

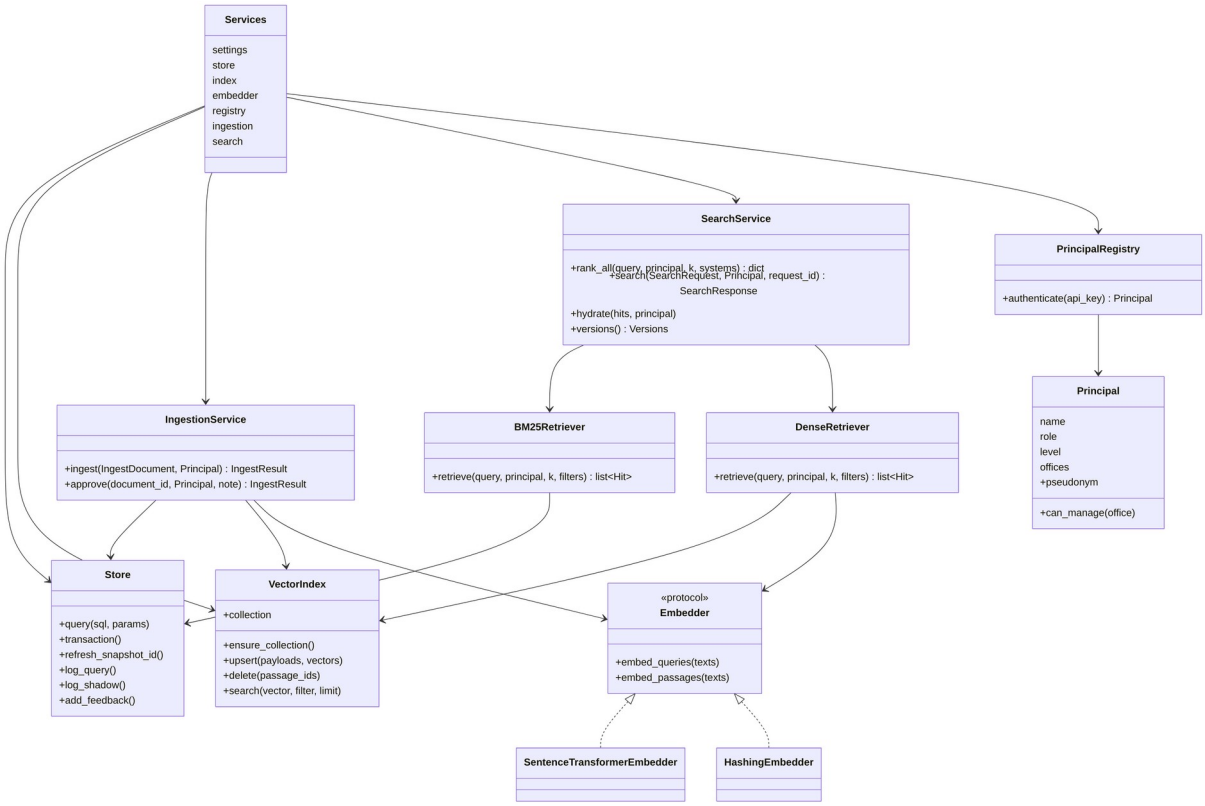


Figure 5. Main classes (names as in the code).

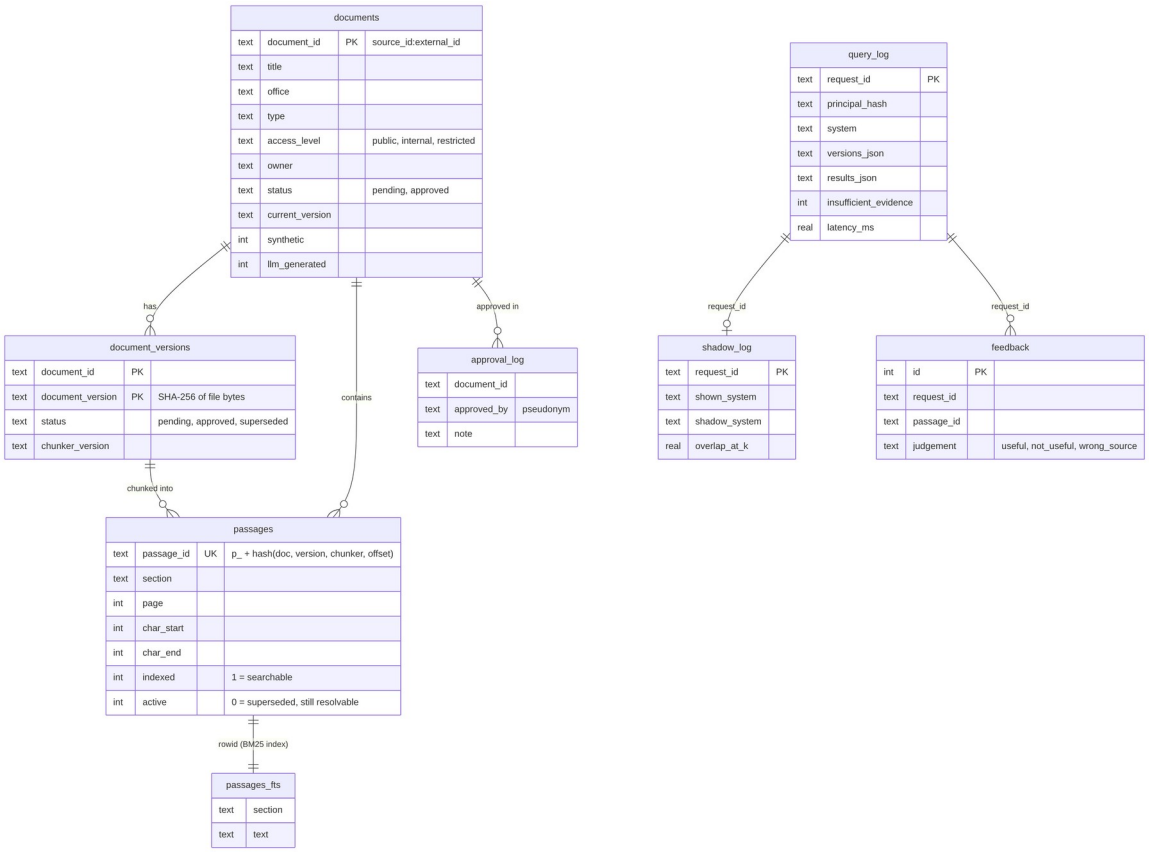


Figure 6. SQLite data model. Qdrant holds one vector per indexed passage.

7. Evaluation

7.1 Method

1. **Question set.** 110 queries (keyword-like, paraphrased, conceptual, out-of-scope; some in English), with graded relevance labels at passage level. The queries and labels were **drafted by an LLM**; see the threats to validity in Section 11.
2. **Split and freeze.** 40 tuning queries and 70 test queries. The test split was frozen with a SHA-256 hash **before** choosing the embedding model and calibrating thresholds.
3. **Systems.** B0 = BM25 only (the keyword baseline); S1 = dense only; S2 = hybrid with RRF. A reranker (S3) was planned as a stretch goal and not implemented.
4. **Metrics.** Recall@K, Precision@K, MRR@10, nDCG@10, correct source in the top 3, correct abstention on out-of-scope queries, access-control leakage, latency.
5. **Targets.** Fixed before running the test set. They are **our own targets**, not official course requirements.

7.2 Results

Table 5. Main results on the frozen test set (source: eval/results/eval_latest.json).

Metric (D1 test: 60 in-scope, 10 out-of-scope)	B0 BM25	S1 dense	S2 hybrid
nDCG@10	0.453	0.603	0.604
Recall@10	0.557	0.722	0.760
Recall@5	0.515	0.700	0.699
MRR@10	0.511	0.642	0.636
Precision@1	0.450	0.533	0.517

Metric (D1 test: 60 in-scope, 10 out-of-scope)	B0 BM25	S1 dense	S2 hybrid
Correct source in top 3	0.583	0.733	0.767
Correct abstention (10 out-of-scope queries)	0.500	0.700	0.700
False abstention rate (in-scope)	0.200	0.067	0.067

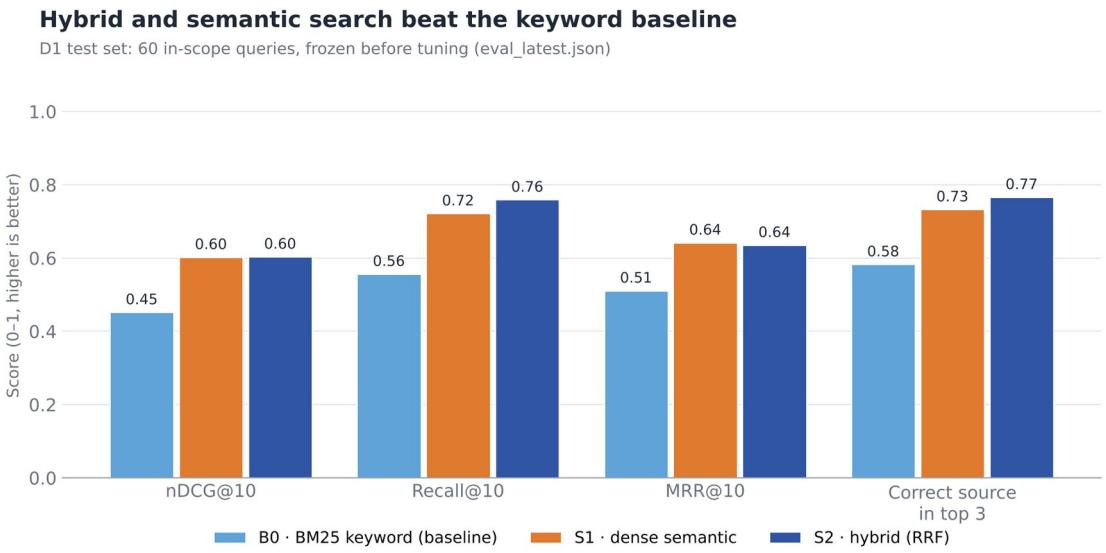


Figure 7. Main ranking metrics on the test set.

Semantic and hybrid search are clearly better than the keyword baseline. Between S1 and S2 the difference on this corpus is small and not statistically significant; S2 helps mostly recall and conceptual queries.

Table 6. Paired sign-flip randomization test (10,000 iterations). Reported as evidence, not used as a success criterion.

Comparison (60 in-scope test queries)	Mean difference	95% bootstrap CI	p (two-sided)
S2 vs B0, nDCG@10	+0.152	[0.089, 0.219]	0.0001
S2 vs B0, Recall@10	+0.203	[0.117, 0.297]	0.0002
S2 vs S1, nDCG@10	+0.001	[−0.051, 0.054]	0.965
S1 vs B0, nDCG@10	+0.151	[0.069, 0.234]	0.0006

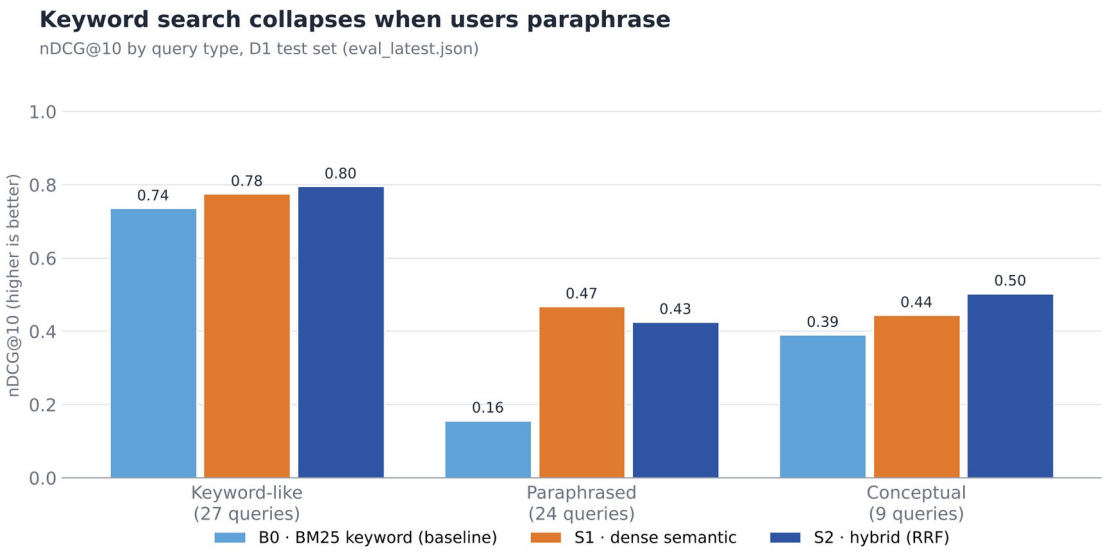


Figure 8. nDCG@10 by query type: keyword search collapses on paraphrased questions.

The most important finding is in Figure 6. On keyword-like queries all three systems work well. On **paraphrased** queries, written in everyday Italian, BM25 drops to nDCG@10 = 0.157, while dense and hybrid retrieval reach 0.468 and 0.426. This is exactly the problem P14 is meant to solve. With English questions over Italian documents BM25 finds nothing, while the multilingual model answers correctly (4 in-scope English test queries, a very small sample).

Table 7. Acceptance targets.

Our target (not official)	Measured (S2)	Met?
T1 · S2 ≥ B0 on nDCG@10 and Recall@10	0.604 / 0.760 vs 0.453 / 0.557	Yes
T2 · Correct source in top 3 ≥ 0.80	0.767	No
T3 · Correct abstention ≥ 0.70	0.700	Yes
T4 · Latency p95 < 1.5 s (no reranker)	77 ms	Yes
T5 · Access-control leakage = 0	0 leaks in 18,576 checks (9 users × 70 queries × 3 systems)	Yes

Latency. Median end-to-end time of a search through the API: 9.4 ms for B0, 46.0 ms for S1, 46.8 ms for S2 (p95: 15.5, 91.1 and 77.0 ms) on one laptop with an RTX 5070 GPU.

Sanity check on D2 (SciFact). On a public benchmark with independent labels, hybrid search is again the best of the three (nDCG@10: B0 0.631, S1 0.659, S2 0.686), as expected. These numbers are not comparable with published BEIR results because the setup is different.

7.3 Error analysis

The failures of S2 on the test set fall into a few groups (full list in docs/EVALUATION.md):

- **Vocabulary mismatch** (8 queries): the answer uses legal terms that do not appear in the question.
- **Right document, wrong section** (2 queries): the citation is close but not precise.
- **Incomplete labels** (3 queries): an equivalent correct answer exists elsewhere but was not labelled.
- **Threshold too strict for English** (3 queries): the evidence threshold was calibrated on mostly Italian queries.

- **Out-of-scope not flagged** (2 queries) and one questionable label.

A cross-encoder reranker and a per-language threshold are the most promising improvements.

8. Demonstration

A Streamlit interface lets the user choose who is searching (the principal), ask a question, and see an extractive summary, the ranked documents with their citations, the evidence of each passage and feedback buttons. The screenshots below come from the running system.

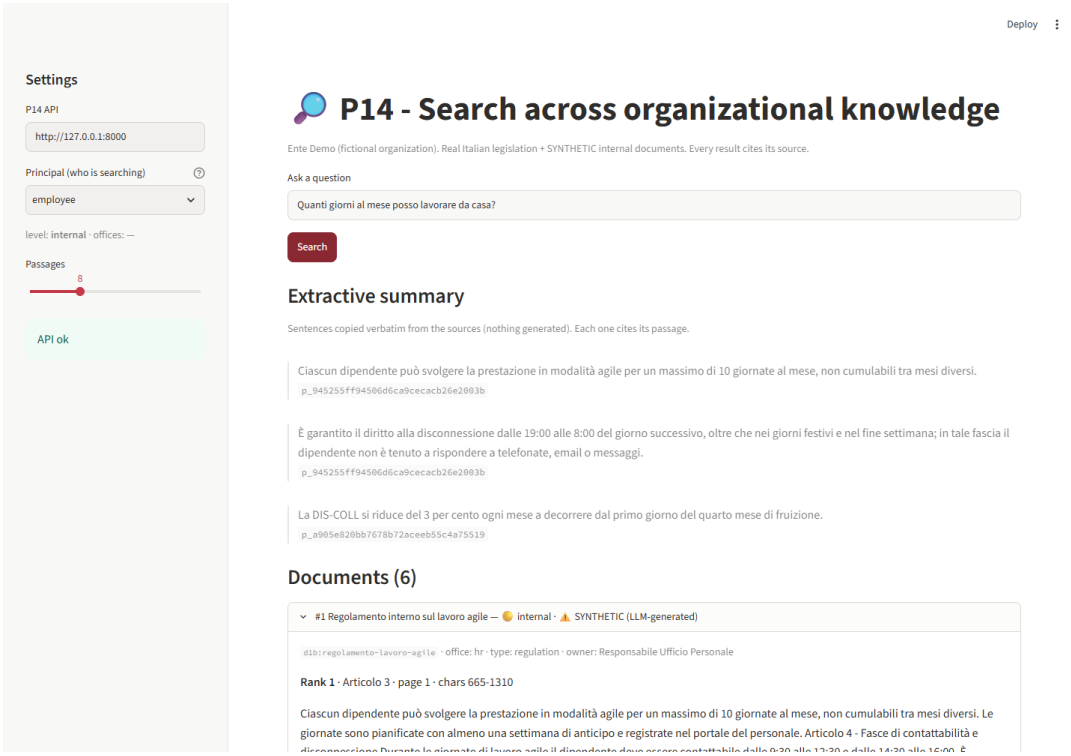


Figure 9. An employee asks "How many days a month can I work from home?" (in Italian). The internal smart-working regulation is ranked first, with its exact article and passage id.

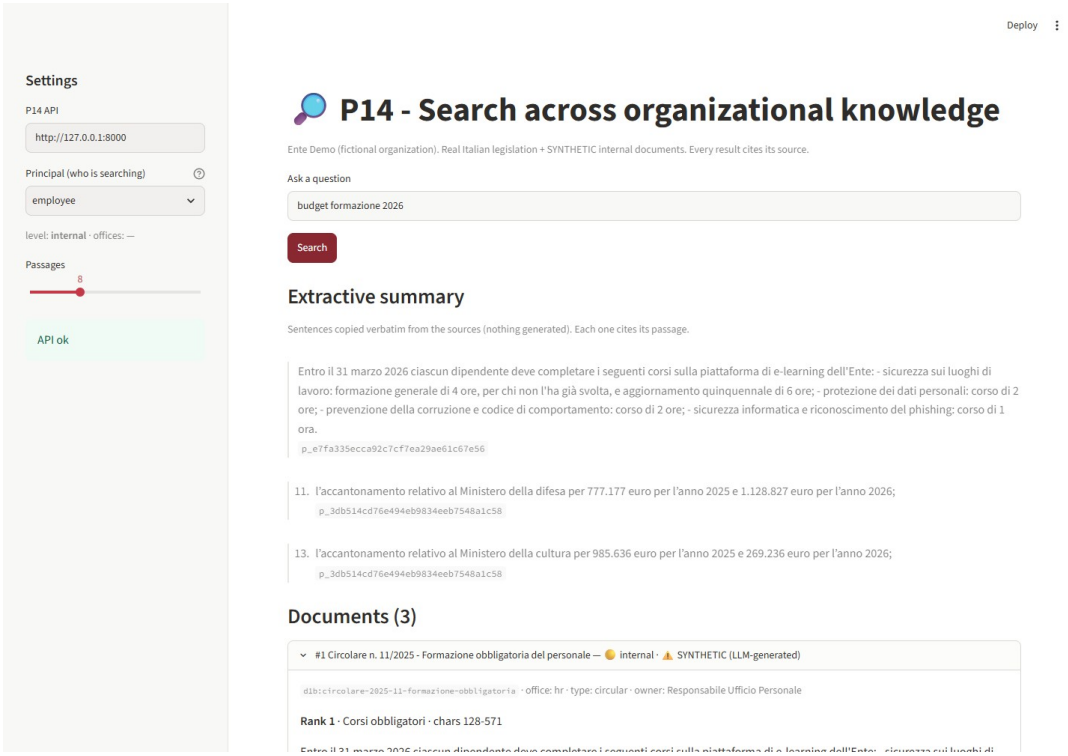


Figure 10. Access control: an employee asks about the 2026 training budget and does not see the restricted budget document.

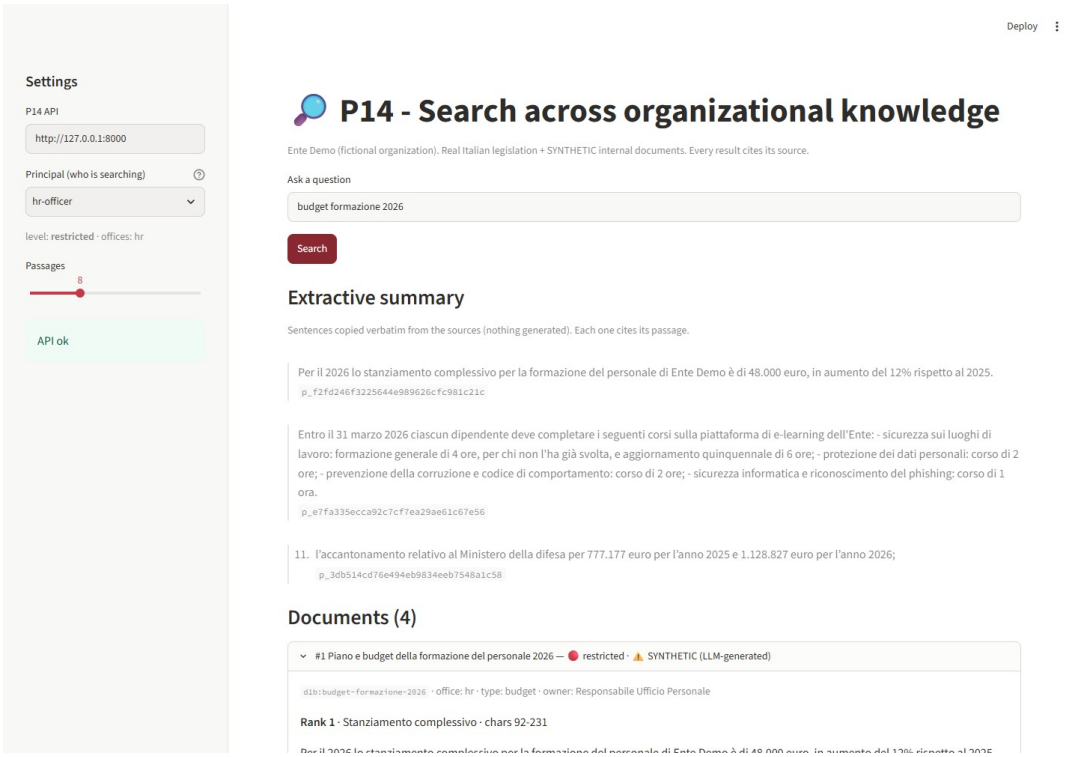


Figure 11. The same question asked by the HR officer: the restricted budget document is now ranked first.

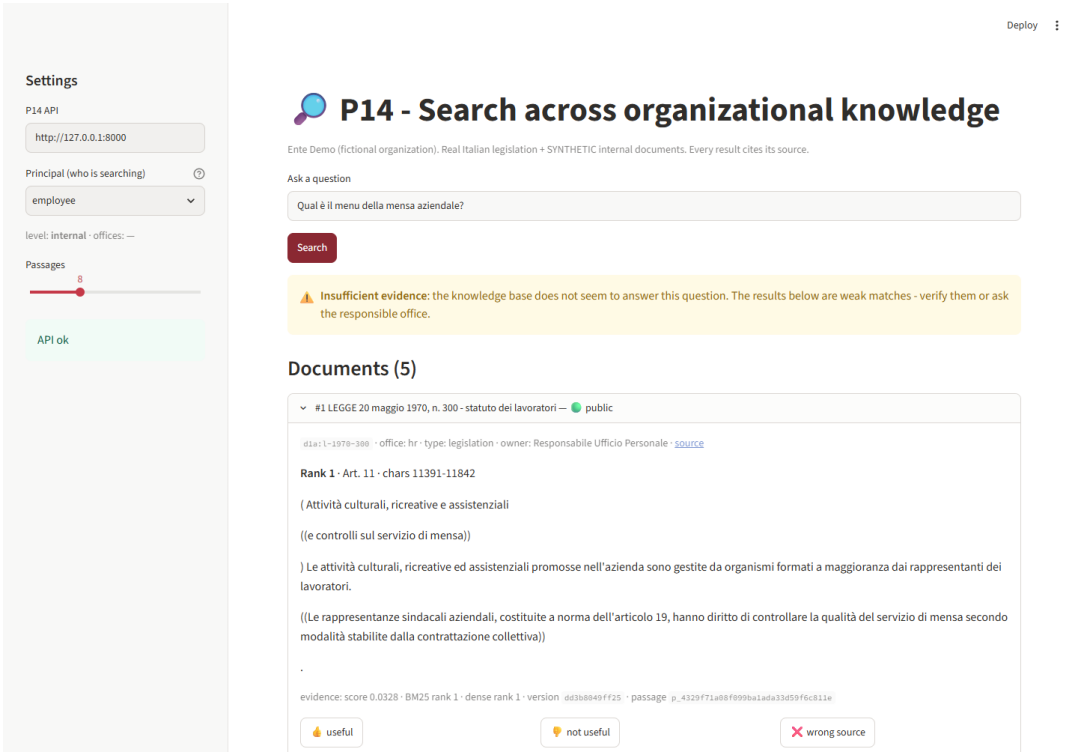


Figure 12. An out-of-scope question (the canteen menu): the system warns that the knowledge base does not answer it instead of guessing.

9. Integration with other puzzle pieces

The contract is a versioned REST API described by OpenAPI (`contracts/openapi/p14-search.yaml`). JSON Schemas in `contracts/schemas/` are the source of truth and an automated test fails if the running API drifts from them. Real examples produced by the running system are in `contracts/examples/`.

Table 8. Main endpoints of the P14 contract.

Endpoint	Purpose
POST /v1/search	Ranked documents and passages with citations, evidence, insufficient_evidence flag and versions
POST /v1/documents · /approve	Ingest an IngestDocument; owner approval makes it searchable
GET /v1/documents/{id} · /v1/passages/{id}	Resolve a document or a cited passage again
POST /v1/feedback	Store the user's judgement on a passage (the operator's final decision)
GET /v1/health · /v1/version	Status and the exact pipeline, chunker, model and index versions

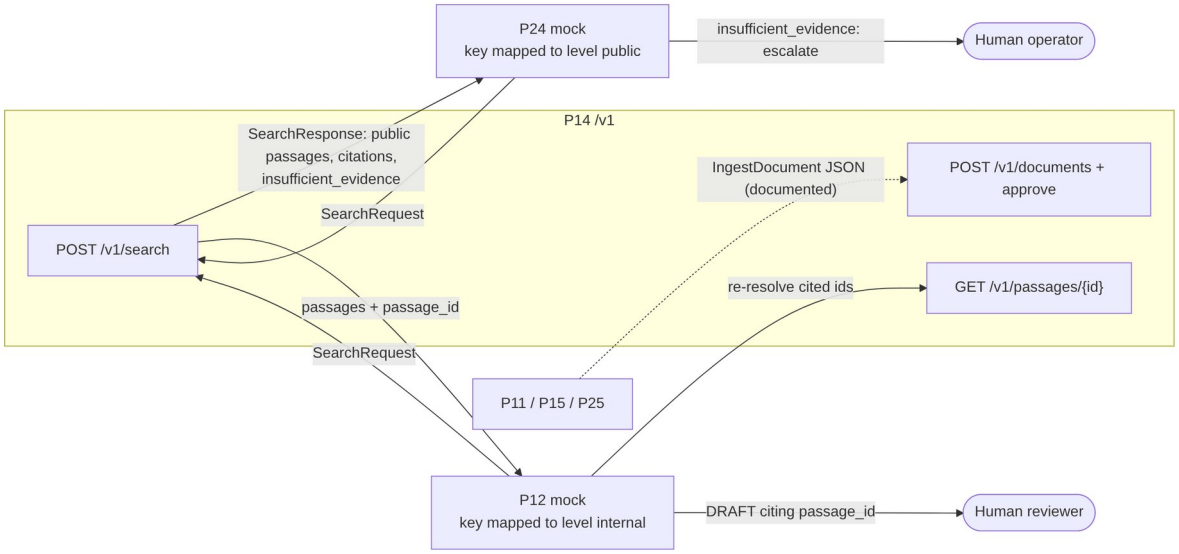


Figure 13. Integration contract and connections.

Table 9. Connections with other AI Puzzle pieces.

Piece	Direction	Data exchanged	Status
P24 External assistant	P14 → P24	Public passages with citations; P24 escalates to a person when evidence is insufficient	Mock implemented and tested
P12 Document generation	P14 → P12	Passages with passage_id, re-resolved before use in a draft	Mock implemented and tested
P11 Document reading	P11 → P14	Structured documents as IngestDocument	Documented
P15 Meeting memory	P15 → P14	Minutes and decisions as IngestDocument	Documented
P25 Expert knowledge	P25 ↔ P14	Validated knowledge cards; retrieval with attribution	Documented (direction to confirm)

An honest observation from the live mocks: P24 correctly escalates unanswerable questions, but for one question about a restricted budget it answered from an unrelated public passage. Access control held (nothing restricted was shown), but the evidence check did not catch the weak match. This is a concrete argument for human escalation and for a reranker.

10. Security, traceability, human oversight and shadow mode

The P14 sheet has no explicit bullet on shadow mode or human oversight, but the framework applies these cross-cutting requirements to every project.

- **Human oversight.** An owner approves every document and its access level; the user always sees citations and decides; weak evidence is flagged; the summary is extractive; feedback stores the final human judgement.
- **Traceability.** Every response carries the pipeline, chunker, model and index-snapshot versions. Every query is logged with a pseudonymised user, the returned passages, scores, latency and feedback.
- **Security.** One API key per consumer (secrets only in `.env`, never in the repository); permission filter inside retrieval with 0 leaks in the evaluation; safe parsing; RFC 9457 error messages without internal details.

- **Validation before automation.** Shadow mode is implemented: the system shown to the user and a shadow system run on the same query and both rankings are logged. A real shadow phase with real users was not possible, so it is simulated by the offline evaluation.

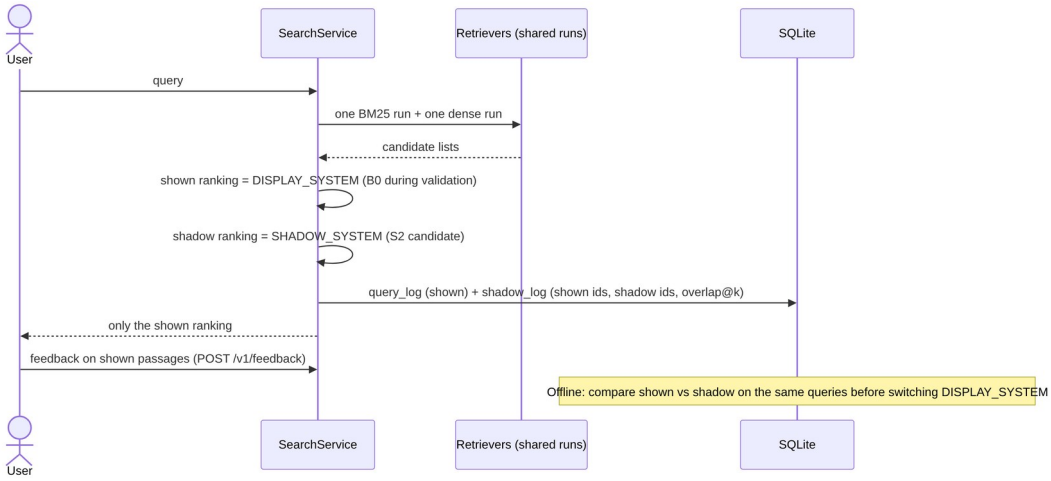


Figure 14. Shadow mode: the shadow system is logged, never shown.

11. Limitations and threats to validity

- **No real organization.** The internal documents are synthetic; access levels of public laws were assigned by us.
- **LLM-drafted evaluation data.** Queries and relevance labels were drafted by an LLM, the same one that wrote the synthetic documents. This can inflate scores through shared wording, and some labels are incomplete. A review by independent human annotators is the main missing step.
- **Small test set.** 60 in-scope and 10 out-of-scope test queries; thresholds calibrated on only 6 out-of-scope tuning queries.
- **Not measured.** The human-judged quality of sources and the real time saved by users.
- **Technical.** No Italian stemming in BM25; some Gazzetta PDF pages remain noisy; API-key authentication only; latency measured on one machine.

12. Future work

- A cross-encoder reranker (S3) and per-language threshold calibration.
- Pooled relevance judgments by independent human annotators.
- A real shadow phase with users of a real organization.
- A related-documents endpoint, CloudEvents on ingestion and an MCP wrapper for LLM-based consumers such as P12 and P24.

13. Conclusion

P14 shows that a small, fully local system can search organizational knowledge by meaning while keeping every answer verifiable and every permission respected. On paraphrased questions, the case that matters most to real users, semantic retrieval roughly triples the quality of keyword search. The evaluation is honest about what is not yet good enough: the correct-source target was missed and the evaluation data still needs independent human review. Because the service is exposed through a stable, tested contract with persistent identifiers, it is ready to become a building block in the final composition of the AI Puzzle.

Appendix A. How to run the project

Requirements: Windows 11, Docker Desktop, Python 3.11; an NVIDIA GPU is optional (set `EMBEDDING_DEVICE=cpu` otherwise). Internet is needed once to download packages, models and documents. Full instructions: README.md and docs/REPRODUCIBILITY.md.

```
python -m pip install --user uv
python -m uv venv --python 3.11 .venv
python -m uv pip install --python .venv\Scripts\python.exe -e ".[dev,data,ui]"
copy .env.example .env
docker compose up -d
.venv\Scripts\python.exe scripts\reproduce.py --clean
```

The last command downloads the corpus, ingests it, runs the tests and the full evaluation, and regenerates every result table (about 7.5 minutes on the test machine). To start the API and the demo: see README.md, section 5.

Appendix B. Repository structure

Table 10. Repository structure.

Folder	Content
src/p14_search/	The service: api, ingestion, indexing, retrieval, security, evaluation, observability
frontend/	Streamlit demo interface
contracts/	OpenAPI spec, JSON Schemas, real input/output examples
mocks/	P24 and P12 mock consumers
data/	Manifest of real documents and sources of the synthetic documents (raw files are downloaded by script)
eval/	Question set, labels, evaluation scripts and all result files
docs/	Architecture, data, evaluation, integration, security, decisions, diagrams, evidence
tests/	80 automated tests

Appendix C. Use of AI tools

AI tools (Claude) were used during this project as an assistant for drafting documentation, writing code, generating the synthetic internal documents and drafting the evaluation queries and labels. All AI-generated data is labelled as such in the repository (`synthetic`, `llm_generated`). Design decisions, the interpretation of results and this report were reviewed by the author, who takes responsibility for them.